



Sustainable development and resources extraction: A novel perspective for resources rich economies

Bo Zhang^a, Meiyu Zhao^b, Yongqian Tu^{c,*}

^a Oriental Digital Technology (Beijing) Co., Ltd, Beijing, 100016, PR China

^b Everbright Prestige Capital Asset Management Co., Ltd, Beijing, 100034, PR China

^c National Academy of Development and Strategies, Renmin University of China, Beijing, 100872, China

ARTICLE INFO

Keywords:

Sustainable development
Resources extraction
Dutch disease
Non-normality
Mineral resources

ABSTRACT

The link between resources and sustainable development has not been explored, especially in resource-rich economies. Natural resource rents are viewed as a source of additional revenue for governments. There are still contradictions in the economic effect of natural resources, which can be beneficial in certain economies but detrimental in others. The current study re-examined the link between natural resources and the economy's growth in the BRICS countries from 1990 to 2021 by assessing the effect of mineral, forest rents and financial market risk. The study investigates the long-run co-integration of the study variables utilizing panel data specifications. Throughout the quantiles, mineral resources adversely influence economic performance, while forest rents have an imbalanced impact on regional economic performance. Financial market risk, on the other hand, has no detrimental influence on the selected countries' economic performance. This research recommends using natural resources more responsibly, increasing renewables, and reducing financial market risk to achieve sustainable development.

1. Introduction

According to the well-known Dutch hypothesis, resource-abundant economies exhibit lower growth rates than those economies with fewer natural resources (NR). Numerous investigations have referred to it as a curse for the country. A rise in remittances strengthens the domestic currency but harms the trade competitiveness and pricey exports of the region. Therefore, due to the Dutch phenomenon, remittances significantly influence the exchange rate, which deteriorates growth correspondingly (Umar et al., 2021a; Yang et al., 2021; Rahim et al., 2021). In simpler words, 'Dutch theory' is the exploitation of NR, leading to the (negative) deterioration of the nation's currency (Chen, 2022). However, NR is an essential and intriguing element in terms of academic research. In the extensive body of literature, the association between NR and economic performance (ECOPER) gave contradictory results. Several objectify that the proper management of the revenues generated from NR could be useful for production, which is beneficial for the country's ECOPER. However, the direction of impact depends entirely on other underlying reasons, such as institutional factors that impact performance alongside NR and could adversely impact economic development (Lashitew and Werker, 2020).

Apart from that, the role of financial variables has risen since the financial crisis (2008), but still, there is infrequent pragmatic literature available in this regard. Financial risk and NR in obtaining economic development have become interesting areas to explore for sustainability. The financial sector's development significantly impacts economic progression and sustainability by increasing several economic opportunities (Hmaitane et al., 2019; Ielasi et al., 2018; Khan et al., 2021; Purewal and Haini, 2022). Moreover, countries exploiting NR and employing financial services can turn the resource curse into a benediction to a nation (Shahbaz et al., 2018). In the prevailing literature, the association between NR and ECOPER is extensive, but the importance of financial risk in obtaining sustainable economic progress is limited. Therefore, based on the needs of the research, it intends to accomplish the following set of objectives. The primary purpose is to revisit the influence of risk from the financial market on ECOPER in the case of BRICS nations; the secondary aim is to evaluate the role of forest rents and mineral rents on the ECOPER in BRICS again.

To accomplish these objectives, the study employs the following variables: ECOPER is employed for ECOPER measured at constant US 2015 dollar prices, FRENT is used for forest rents, MRENTS is for mineral resources, and lastly, FRI is employed for financial market risk for the

* Corresponding author.

E-mail addresses: zhangbo@or-intech.com (B. Zhang), zhaomy@ebpcapital.com (M. Zhao), stevetu@ruc.edu.cn (Y. Tu).

<https://doi.org/10.1016/j.resourpol.2023.103595>

Received 28 December 2022; Received in revised form 30 March 2023; Accepted 13 April 2023

Available online 10 May 2023

0301-4207/© 2023 Elsevier Ltd. All rights reserved.

sample panel data economies from 1990 to the year 2021. In addition, up-to-date co-integration analysis and Quantile regressions econometric techniques are applied to investigate the abovementioned objectives. Attributable to non-normal distribution, the study employs Quantile regressions for unbiased and reliable results.

The study is important in re-examining the role of revenues from NR on ECOPER in BRICS.¹ Since BRICS is imperative to the world's economy, which highly contributes to 25% of the nominal GDP globally, it has an enormous value in world trade and the forex market (Chen et al., 2023; Khan et al., 2020, 2023; Mirza et al., 2023). BRICS countries significantly contribute to economic prosperity by benefiting the population countries. Besides, its impact on the regional and world economy has steadily increased since its inception. Therefore, the current study is significant in assessing the role of resource rents and financial risk on ECOPER in BRICS by providing noteworthy conclusions for evaluating sustainability policies. All group economies working under a shared vision bring fruition to the world economy socially, politically, economically, and financially. The lesser the country extracts NR, it substantially promotes environmental sustainability by reducing environmental difficulties caused by using NR, such as natural gas, oil, coal, and so on. Besides, this study advises strengthening financial institutions by creating checks and balances, decreasing corruption, and channeling funds.

The study subsidizes the literature in the following ways. Firstly, the study provides additional evidence in the empirical literature on the connection between revenues from resource rents and ECOPER. The study uses different resource indicators to analyze NR on ECOPER comprehensively. The literature is extensive with contradictory results; therefore, the present study findings/conclusions provide an additional suggestion for evaluating policies besides extending the long debate on resource rents and economic development. Second, the impact of financial risk on ECOPER is empirically observed. The literature is scarce in this area; therefore, the research intends to contribute the study and its findings to the empirical literature in the case of BRICS nations. Third, the study empirically explores resource rent indicators on ECOPER in BRICS, which is crucial because the overall resource rents provide the average influence of resources. However, the current study's purpose of scrutinizing the importance of MRNT and FRNT, which are relatively scant in the existing literature, particularly in the countries under discussion. Fourth, the study examines the combined nexus between risk from the Financial Market, Resource rents, and ECOPER in BRICS countries. For this, the study uses updated data from 1990 to 2021 and employs updated econometric analysis. The increasing economic potential in these nations influences the world globally and its international systems and organizations for policy frameworks. Therefore, the new findings will deliver compact suggestions for researchers, economists, and policy regulators to re-vamp economic and environmental policies.

The remaining manuscript is organized as follows. The upcoming section after the introduction deals with the literature assessment of the study factors. Section 3 and section 4 are about the model, methodology, results, and economic interpretation, respectively. Whereas Section 5 presents the conclusion and relevant policy suggestions.

2. Review of literature

This segment presents some relevant pragmatic literature on the nexus of dependent and independent variables for research analysis. Therefore, section 2 is divided into two sub-sections to evaluate the role of NR, forest rents, and financial market risk on ECOPER.

2.1. Role of forest rents and NR in ECOPER

The NR is a source of capital formation in an economy if efficiently managed. It aids in limiting poverty and increasing income, besides creating jobs and national wealth (OECD, 2011). However, in the extant literature, there exist negative and positive influences of NR on the ECOPER of the economy. It could be a source of blessing for a nation or become a curse by deteriorating economic development. For instance, Zhou et al. (2022) examined both impacts on ECOPER. The empirical results demonstrated that NR and forest rents reduce the ECOPER of the South Asian economy. In contrast, oil, coal, and natural gas rent tend to increase ECOPER the economy. All variables show a bidirectional causal relationship, excluding the relationship between coal rents and ECOPER. Similarly, Yilanci et al. (2021) scrutinized the mixed to no influence of NR on ECOPER depending on the country's income level. Correspondingly, Ampofo et al. (2020) inspected the asymmetric impact on NR and ECOPER in panel data of mineral-rich countries. Some economies showed a positive association between NR and ECOPER, while others demonstrated a negative relationship. In another innovative contribution to the literature, Xie et al. (2022) analyzed the influence of forest rents on ECOPER by employing global statistics. The empirical results disclosed one-way causality between forest rents to ECOPER globally. Further, in another recent analysis of NR and ECOPER nodes, Wen et al. (2022) observed the encouraging role of total NR on the economy's performance in BRICS with bidirectional causality, i.e., increasing rents increases the ECOPER. In the case of India and Pakistan, Mehar et al. (2018) assessed the nexus between ECOPER and NR from 1970 to 2017. The pragmatic findings confirmed the positive impact of NR on ECOPER, leading to significant economic development in the economy. Likewise, in BRICS nations, Deng et al. (2022) inspected the nexus between NR and ECOPER and found a positive association demonstrating that NR is a blessing to the nation with uni-directional causality. In dissimilarity, Shabbir et al. (2020) inspected the harmful impact of total NR on the ECOPER of the region. In the former study on the nexus between rents of NR and ECOPER, Jović et al. (2016) investigated the influence of forest rents on ECOPER. The findings depicted the major impact of forest rents on ECOPER. Likewise, in another research, Ofori and Grechyna (2021) reviewed the optimistic connection between forest rents and ECOPER. Huang et al. (2020) established that forest rents have momentarily contributed to ECOPER in Asian economies. In Tunisia, Kwakwa et al. (2021) studied the positive association of forest rents with the ECOPER of the country. In addition, Mohamed (2020) claimed the positive relationship between NR, such as oil and forest rents, on ECOPER, which ultimately becomes negative in the long run with an uni-directional connection from rents to economic growth. In contrast, Inuwa et al. (2022) studied the undesirable impact of forest revenue on ECOPER, verifying the resource curse hypothesis in Nigeria from 1981 to 2017. In the case of West African economies, Asiedu et al. (2021) confirmed the inverse effects of forest revenue on ECOPER. Increasing the revenues on forest rents negatively impacts the economy's performance.

2.2. Nexus between financial market risk and ECOPER

After the financial crisis (2008), the financial risk influence on ECOPER seems very important. Several authors emphasized market risk's impact on ECOPER (Ahmed, 2020; Bansal and Kumar, 2021; Su et al., 2022; Umar et al., 2021b). However, in the present academic literature, the association between financial market risk and ECOPER is quite infrequent. Nonetheless, this sub-section elaborates on the nexus to exemplify the aspects of the relationship for research analysis. Ekinci (2016) explored the relationship between financial market risk and the performance of banks in the Turkish economy. The empirical findings demonstrated that market risk significantly impacts the bank's performance, which significantly affects the ECOPER. In a recent study on the case of OECD economies, Ahmad et al. (2022) scrutinized the causal

¹ Where B: Brazil; R: Russia; I: India; C: China; and S: South Africa.

relationship between financial risk and sustainable development. The findings showed a bi-directional nexus between financial market risk and sustainability and vice versa, depicting that enhancing financial stability reduces financial risk, which is imperative for economic sustainability and performance through improved decision-making processes and economic analysis (Pan et al., 2021). In European Union nations, Jianu et al. (2020) examined the country's significant association between financial risk and ECOPER. Gadea Rivas, Laeven, and Pérez-Quirós (2020) emphasized that increasing financial risk deteriorates ECOPER by negatively impacting economic development. However, using macroprudential policies could be useful in managing risk and growth trade-offs. For sub-Saharan Africa, Ibrahim and Alagidede (2022) Ibrahim and Alagidede (2018) highlighted that the improvement in financial development is significant in improving the ECOPER of the country and promoting development. Further, it reduces financial risks in the economy. In analyzing financial institutions and market impact on ECOPER, Purewal and Haini (2022) employed these financial measures to capture the wholesome impact of finance on economic sustainability. The experimental results showed a U-shaped relationship between financial variables and ECOPER.

3. Data and methods

3.1. Data and model description

Due to a growing understanding of resource extraction's economic, social, and ecological implications, resource-rich countries must investigate sustainable development and resource extraction. During the last three decades, it has become clear that conventional resource extraction methods frequently hurt the environment and populations in resource-rich destinations. Since global NR demand rises, sustainable resource management is increasingly important. Unsustainable resource exploitation may damage the environment, cause social unrest, and destabilize resource-rich economies' economic expansion and sustainability, particularly in the longer term. In contemporary times, the existing policies regarding the management of NR for sustainable economic growth must be revised as some of the NR components (such as minerals) are harming economic growth. Therefore, it is a need of time to empirically examine the association between these variables for constructing and developing relevant and appropriate policies.

From the given literature and objectives of the study, the present research opted for the gross domestic product (GDP) to capture the ECOPER of the region. Nonetheless, the GDP reflects the economy's healthiness by considering aggregate investment, consumption, government expenditure, etc. Consequently, the ECOPER could be appropriately represented via GDP since the study objectives are to analyze the connection between NR, financial market risks, and ECOPER. Therefore, this study uses mineral rents (MRENTS), forest rents (FRENT), and financial risk index (FRI) as the key explanatory factors in the model. Data for the discussed variables are obtained from several

sources covering the period from 1990 to 2021 for the BRICS region. The specifications and data sources are presented in Table 1.

Following the latest research of Wen et al. (2022), Jianu et al. (2020), and Inuwa et al. (2022), this research develops the study model given below:

$$ECOPER_{it} = f(MNRENTS_{it}, FRENT_{it}, FRI_{it})$$

The general model demonstrates that ECOPER is the function of mineral rents, forest rents, and financial market risks. The discussed general model could be transformed into the following econometric model:

$$ECOPER_{it} = \alpha_1 + \gamma_1 MRENTS_{it} + \gamma_2 FRENT_{it} + \gamma_3 FRI_{it} + \varepsilon_{it} \quad (1)$$

In the econometric model, the α stands for the model's intercept, γ 's represents the slope of each regressor, i and t reveal the cross-section and time, while ε captures the model's random error term.

3.2. Estimation strategy

Before predicting the coefficient values for the long run, this research synthesizes the information by assessing descriptive stats for each of the model's components. The descriptive statistics particularly contain the median, mean, and range values. However, the ranging value is included in the period's highest and minimum values, which vary significantly. The differentiation between these variables reveals the variations or instability of the information. The standard deviation, a common measure of a variable's volatility, determines each variable's volatility in this research. In addition, this research employs other metrics, such as Kurtosis and skewness, to evaluate the data's normalcy since the issue of non-normality results in a biased estimate when evaluating long-run elasticities. This study also employs the conventional Jarque and Bera (JB) (1987) normality test, which is more robust than the previous measures because it evaluates both the excess Kurtosis and skewness to demonstrate the variables' regularity. The subsequent formulation could be used to evaluate the testing in inquiry:

$$JB = \frac{N}{6} \left(S^2 + \frac{(K-3)^2}{4} \right) \quad (2)$$

The JB test suggests that each variable's data is regularly spread by asserting that the excess Kurtosis and skewness are equal to zero. Nevertheless, if the calculated value surpasses the critical values, the test's assumption will be neglected, and the conclusion will be that the variables have an asymmetrical distribution.

After evaluating the normality, this research scrutinizes the variables' stationarity. Since unit root testing is crucial for examining long-run estimates, therefore, this study uses a set of five unit root estimators, which includes Breitung (2001), Im et al. (1997), ADF-Fisher (Maddala and Wu, 1999), Levin et al. (2002), and Phillip-Perron (Phillips and Perron, 1988). All of these tests are employed on the leveled and first differenced data to offer a clear picture of the variables' stationarity. The tests mentioned above reveal the existence of a unit root in a panel time series as the null hypothesis. However, the null prediction may be rejected after obtaining higher statistical values than their critical values at 1, 5, and/or 10 percent levels.

Once the stationarity of the variables is verified, this research examines the long-run connection of the study variables. The present study uses the Kao (1999) residual co-integration test. This technique seeks to identify causal relationships between variables by analyzing if the unpredictable patterns of a set of variables are shared across the series. If both trends are similar, either one variable is the source of the other, or a third variable influence both. The null supposition of Kao's assessment is the existence of the long-run association between the study variables until significant statistical values are obtained.

This research empirically investigates the long-run effect of the components under discussion on ECOPER by utilizing the quantile

Table 1
Specification of variables and data source.

Variable	Specification	Data Source
ECOPER	Economic performance is considered as the GDP at constant US\$ 2015 prices.	https://databank.world-bank.org/source/world-development-indicators
MRENTS FRENT	Sum of mineral rents, % of GDP The Roundwood harvest multiplied by the goods of domestic pricing and a domestic rental rate determines forest rents – measured as a GDP percentage	
FRI	Financial market risk is the risk of financial investment losses due to unfavorable price fluctuations.	https://www.prsgroup.com/

regression assessment discovered by [Koenker and Bassett Jr. \(1978\)](#). This technique is efficient here since the test estimates from [Jarque and Bera \(1987\)](#) reveal an unbalanced distribution, implying that standard methodologies cannot produce correct values. In order to avoid the overestimation and underestimating biases of the calculated coefficients in these conventional methods, the quantile regression – which delivers the projected estimates at each defined quantile, was used in this study. The quantile regression methodology is superior to the usual least square method since it accounts for distributional and individual variability, giving full information about the association between the research components ([Cheng et al., 2019](#)). Moreover, the quantile regression is also more resilient than standard regression(s), which provides the average contribution of explanatory variables ([Qin et al., 2021](#)). The previously presented regression equation, namely Eq. (1), may be translated into the quantile regression format, as demonstrated in the following equation:

$$Q_{ECOPER_{it}}(\theta | \alpha_i, \varphi_i, X_{it}) = \alpha_i + \varphi_i + \varphi_{1,\theta} MRENTS_{it} + \varphi_{2,\theta} FRENT_{it} + \varphi_{3,\theta} FRI_{it} + \varepsilon_{it} \quad (3)$$

The current research adopted four quantiles, namely Q25th, Q50th, Q75th, and Q90th, to scientifically investigate the effects of MRENTS, FRENT, and FRI on ECOPER of the BRICS regions, in which the above equation's subscripts reflect the quantile of each study variable.

After collecting empirical data using quantile regression, this research applies a non-parametric method to quantify the model's robustness. The robust predictor used in this investigation is the panel dynamic ordinary least squares (PDOLS). In contrast to quantile regression, this method yields strong average coefficient values for each variable under consideration.

4. Results and discussion

This segment offers the estimated outcomes attained via the specifications given above. Firstly, this study evaluated the descriptive statistics elaborating the median, mean, and range values, as reported in [Table 2](#). In these specifications, the outcomes stated that the values are positive for ECOPER, MRENTS (except for the minimum), and FRI. Whereas the median, mean, and minimum values of FRENT are found to be negative, which also reached the maximum of 0.0429. Nonetheless, the minimal and maximal values of the variables hold substantial differences, which forces this research to assess the components' standard deviation. Since the standard deviation also provides volatility of each variable, which is found highest in ECOPER, followed by MRENTS, FRENT, and FRI. Apart from the mentioned specifications, this study also evaluated the variables' normality. Accordingly, this study evaluated the skewness and Kurtosis for each variable given. These two normality specifications offer the non-normal spread of the variables' series. Nonetheless, the measures indicate normality; still, this study uses the JB test of normality, which considers both the excess Kurtosis and skewness simultaneously. The projected results of the latter test asserted that the variables except ECOPER are statistically significant,

Table 2
Descriptive results.

	ECOPER	FRENT	MRENTS	FRI
Mean	12.06042	−0.372562	0.608545	1.571596
Median	12.07913	−0.331541	0.576230	1.589438
Maximum	13.16530	0.042873	1.342646	1.681241
Minimum	11.25372	−1.071059	−0.022675	1.250217
Std. Dev.	0.449237	0.269256	0.320235	0.082066
Skew.	0.348025	−0.656262	0.429621	−0.972669
Kurt.	3.100451	2.724753	2.589761	3.962186
JB	3.297180	11.98988	6.043953	31.40097
Probability	0.192321	0.002491	0.048705	0.000000

neglecting the null supposition and accomplishing that FRENT, MRENTS, and FRI follow the asymmetrical distribution.

After the descriptive stats, the research runs the diagnostic test, which is needed to investigate long-run coefficients. Specifically, the stationarity of the variables is examined by applying five tests of stationarity, i.e., [Breitung \(2001\)](#), [Im et al. \(1997\)](#), ADF-Fisher ([Maddala and Wu, 1999](#)), [Levin et al. \(2002\)](#), and Phillip-Perron ([Phillips and Perron, 1988](#)). The empirical findings of the test are reported in [Table 3](#). From the examined estimates, this study noted that all the specifications provide insignificant results for ECOPER. In contrast, FRENT, MRENTS, and FRI are significant at only 10%, 1%, and 5% levels in the Levin, Lin and Chu, and Breitung tests. However, the statistical estimates are insignificant for the Im, Pesaran, Shin, ADF-Fisher, and Phillip-Perron tests. This study also analyzed each variable's stationarity at I(1), where all the variables are significant at 1, 5, and 10 percent levels. Thus, the null prediction of all the variables is statistically significant and concludes that the variables are stationary, which satisfies the properties for the long-run coefficients estimations.

After the stationarity testing, this study employs the Kao residual test for co-integration, approximations of which are conveyed in [Table 4](#). This study observed that the projected values are significant at 10%. Hence, the test's null proposition may be neglected to conclude that the presence of co-integration between the variables, including ECOPER, FRENT, MRENTS, and FRI, is present, which also satisfies the property for the long-run coefficients examination.

Since the Kao residual co-integration test validates the presence of the long-run connection amid variables under discussion, this research tends to estimate each variable's coefficients in the longer run. Further, the JB normality test affirms the non-linear data distribution. Accordingly, this research uses panel quantile regression which is more powerful than the traditional least square estimates in tackling the issue of non-normality. The outcomes after estimation are described in [Table 5](#). The results estimated reveal that NR rents are adversely associated with the economic prosperity of the BRICS countries. Specifically, the study noted that a percent increase in the MRENTS leads to a drop of ECOPER by 0.380–0.549%, the magnitude of which is found to increase across the quantiles. These experimental outcomes are found to be consistent with the empirical results of [Zhou et al. \(2022\)](#), [Yilanci et al. \(2021\)](#), and [Ampofo et al. \(2020\)](#) by validating the harmful effect of mineral rents on prosperity and growth of the economy.

Table 3
Unit root testing.

Unit Root Tests: Summary [I(0)]				
	ECOPER	FRENT	MRENTS	FRI
Levin, Lin & Chu t^*	2.54870	−1.47413*	1.07767	−1.93022**
Breitung t-stat	4.01457	−1.33020*	−2.73907***	0.67679
Im, Pesaran and Shin W-stat	2.09345	−0.58707	0.56434	0.16273
ADF - Fisher Chi-square	8.55005	11.1813	5.52907	7.25638
PP - Fisher Chi-square	4.52077	8.84093	10.2578	9.57140
I(1)				
Levin, Lin & Chu t^*	−0.75510	−7.33593***	−3.64776***	−7.32451***
Breitung t-stat	2.23103	−6.36636***	−7.55537***	−5.54899***
Im, Pesaran and Shin W-stat	−0.76540	−6.55394***	−6.41411***	−8.31750***
ADF - Fisher Chi-square	16.0942*	55.3312***	54.0023***	71.9550***
PP - Fisher Chi-square	29.7173***	178.346***	188.207***	160.534***

Note: I(0) symbolizes the level's stationarity, while the I(1) is for the first difference. Asterisks **, *, and *** represent the significance level at 10%, 5%, and 1%.

Table 4
Co-integration testing.

Kao Residual Co-integration Test		
	Test	Prob.
ADF	−1.416118*	0.0784
Residual var.	0.000638	
HAC var.	0.001845	

Note: Asterisks ** ** and *** represent the significance level at 10%, 5%, and 1%.

Table 5
Long-run estimates – quantile regression.

Quantile $Q_{0.25}$				
Variable	Coef.	Std. Err.	t-Stat.	Prob.
MRENTS	−0.380071***	0.126865	−2.995875	0.0032
FRENT	−0.347487**	0.170753	−2.035023	0.0435
FRI	7.488051***	0.087497	85.58110	0.0000
Quantile $Q_{0.50}$				
MRENTS	−0.472422***	0.116441	−4.057168	0.0001
FRENT	0.107022	0.158140	0.676756	0.4996
FRI	7.855962***	0.076137	103.1824	0.0000
Quantile $Q_{0.75}$				
MRENTS	−0.434700***	0.135763	−3.201890	0.0017
FRENT	0.381245**	0.192199	1.983594	0.0490
FRI	8.070192***	0.098337	82.06639	0.0000
Quantile $Q_{0.90}$				
MRENTS	−0.549155	0.351755	−1.561187	0.1205
FRENT	1.091542***	0.303073	3.601578	0.0004
FRI	8.583656***	0.218696	39.24931	0.0000

Note: Dep. var. is ECOPER, while asterisks ** ** and *** represent the significance level at 10%, 5%, and 1%.

On the other hand, FRENT is found asymmetrically affects the ECOPER of the study countries. More precisely, a one percent rise in FRENT substantially enhances and reduces ECOPER in the lower ($Q_{0.25}$) quantile by 0.347% while increasing it by 0.381 and 0.092% in the upper ($Q_{0.75}$ – $Q_{0.90}$) quantiles, respectively. Such conclusions are consistent with the observed literature. For instance, Inuwa et al. (2022) illustrate the similar (negative) linking between ECOPER and forest rents, while Jović, Maksimović, and Jovović (2016), Ofori and Grechyna (2021), and Kwakwa et al. (2021) reports the constructive influence of such rents on economic sustainability in different regions, which also backs the results of this study. The frequency of NR use boosts output but increases its depletion rate, affecting economic growth. Manufacturing, which boosts economic growth, requires natural resources. Dependency on NR usage is ineffective for sustainable development due to the reduced quality of NR and the dwindling yield of factor input. Thus, NR overuse harms viability. Transforming NR into goods and services boosts their economic value. The transfer of NR into useable goods and services drives prolonged economic growth, which is the basis of sustainability. Rent-seeking may impede NR use, limiting economic growth along with positive effects. Rents from NR extraction allow unproductive consumption at the cost of actual output, which increases non-tradable sector activities and decreases output. The “Dutch Disease”—poor economic growth due to the lack of “backward and forward” ties throughout economic sectors—results from this.

Contrary to MRENTS and FRENT, this research found a significant positive influence of financial market risk on ECOPER across the selected quantiles at a 1% significance level. More specifically, an enhancement of one percent in the FRI leads to an increase in the ECOPER by 7.488–8.584%, which is found to increase from lower to upper quantiles. Financial concerns can significantly impact the firm, its surroundings, and the government. Considering a scenario in which the

government loses control of its monetary policy, cannot manage inflation, and defaults on its bonds or other debt obligations.² Opposing to the statement, this study validates the positive and substantial impact of FRI on the BRICS economies ECOPER. The main reason behind the constructive linkage between these components is that the BRICS regions are paying appropriate attention to the stability of financial markets, which reduces the risk associated with the financial market risks. Therefore, the ECOPER of these countries is increasing during the study period. This study’s empirical outcomes are in line with the results of Ekinci (2016) and Jianu et al. (2020).

On the other hand, the evaluated outcomes contradict the studies of Ahmad et al. (2022) and Pan et al. (2021) by claiming that the increased financial market risk could harm the economic stability and sustainability of the region. Apart from the coefficients’ value, the variables’ standard errors are found to be minimal, further strengthening the variables’ significance. The coefficients are also portrayed on the graph (see Figure 1).

After estimating the results via quantile regression, this study further established the model’s robustness using the non-parametric approach. Specifically, this research applies the panel dynamic ordinary least square test, and the projections are reported in Table 6. The evaluated results of the PDOLS asserted that MRENTS and FRENT adversely affected the ECOPER. At the same time, FRI is a significant factor in

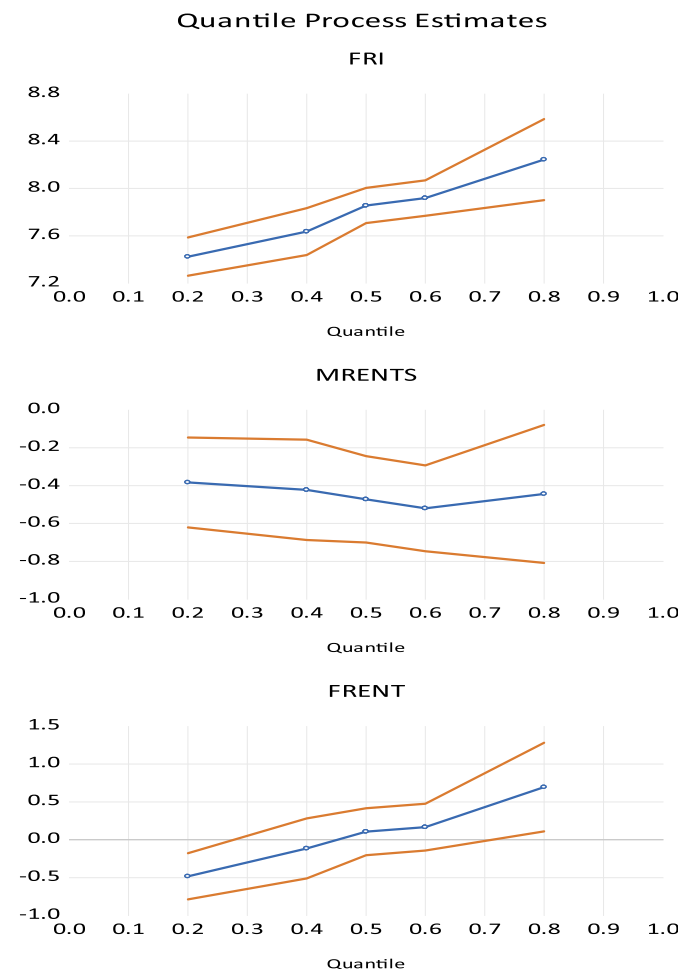


Figure 1. Quantile Estimates (Graphical view).

² See for instance: <https://talentedge.com/blog/impact-financial-risks/>.

Table 6
Robustness testing – non-parametric approach.

Variable	Coef.	Std. Err.	t-Stat.	Prob.
Panel Dynamic Ordinary Least Square				
MRENTS	−0.521199***	0.102126	−5.103489	0.0000
FRENT	−0.849829***	0.130063	−6.533972	0.0000
FRI	0.370295***	0.064715	−5.721934	0.0000
R ²	0.972612			
Adj. R ²	0.957131			

Note: Dep. var. is ECOPER, while asterisks ** *** and *** represent the significance level at 10%, 5%, and 1%.

ECOPER. All the results are significant at the 1% level. It validates the specific influence of each explanatory variable on ECOPER, whereas the higher R² and adj. R² demonstrates the model's goodness.

5. Concluding remarks

5.1. Conclusion

Due to the paradoxes, the linkage between NR and ECOPER is still being debated in the academic literature, with certain researchers claiming NR is a blessing. In contrast, others see it as a curse that leads the economy to Dutch disease. Concerning this, the present study seeks to reinvestigate the BRICS economies for the mentioned nexus and scrutinize the connection between financial market risk and regional ECOPER. The results support the stationarity and durability of the long-run equilibrium variables' association while adopting panel assessment techniques such as unit root testing and co-integration specification. Nonetheless, the problem of data's non-normality requires appropriate estimation techniques to avoid biased conclusions. Therefore, this research applies quantile regression to analyze the influence of each regressor at a particular quantile. The projected results illustrate that NR negatively influences the BRICS nations' ECOPER. These findings confirm the prevalence of the Dutch plague in the BRICS economies. Because these economies are freshly emerging, various issues, particularly poorer institutional quality, do not adequately manage natural resource rents, harming ECOPER and growth.

On the other hand, forest rents are shown to be asymmetrically hurting the region's ECOPER. Nonetheless, the rents from forests stimulate various sectors, enhancing the BRICS economies' performance. Nonetheless, forest loss may jeopardize these economies' economic and environmental viability. On the other hand, financial market risk is positively related to the area's economic success, indicating reduced financial market risk in the BRICS region. Still, there is a need for practical policy implications that might aid in risk reduction, resource sustainability, and ECOPER.

5.2. Policy implications

Based on the empirical findings, this study proposes strategies that might lead to the BRICS economies achieving sustainable development. Policies governing natural resource extraction and its consumption or utilization might be reformed to emphasize the limited extraction of forest and mineral resources and other NR. Specifically, the BRICS economies may adopt diversification in their economic systems. Higher resource extraction could increase a country's dependency on these non-renewable resources. Therefore, these economies shall invest more in other sectors such as agriculture, tourism, and manufacturing. The rents from the forest and mineral resources could be used in infrastructural projects, education, human capital, research and development, and technological innovation. Investment in these sectors could significantly reduce the dependency level of NR in these economies, promoting economic sustainability. Furthermore, this study advises that financial institutions be strengthened by decreasing corruption and channeling

funds to reduce corruption and mismanagement of funds, which promote risk. Authorities must act in this sector to lower the risk connected with financial markets to boost investment in various industrial and economic initiatives.

5.3. Limitations and future research suggestions

The study has a few limitations upon which it proposes future recommendations. The study uses limited resource indicators for evaluating the role of resource rents on ECOPER, which is a major limitation of the study. Therefore, it is suggested to explore all the present resource indicators on ECOPER for a comprehensive analysis of resource rents which helps formulate economic policies in the future. Second, the existing literature comprehensively analyzes variables such as institutional and natural resources. However, then again, empirically exploring this in future research is suggested. Third, data unavailability is another study limitation. Thus, it is recommended to examine these with updated data periods with the inclusion of other relevant policy variables. It will provide additional evidence for sustainability areas for assessing sustainable policies.

Author statement

Bo Zhang: Original Idea; Estimation
Meiyu Zhao: Writeup; Review
Yongqian Tu: Revision; Overall draft improvement
Sustainable development and resources extraction: A novel perspective for resources rich economies

Declaration of competing interest

No competing interest reported by any of the author.

Data availability

Data will be made available on request.

References

- Ahmad, M., Ahmed, Z., Gavurova, B., Oláh, J., 2022. Financial risk, renewable energy technology budgets, and environmental sustainability: is going green possible? *Front. Environ. Sci.* 518.
- Ahmed, W.M., 2020. Asymmetric impact of exchange rate changes on stock returns: evidence of two de facto regimes. *Rev. Account. Finance* 19, 147–173.
- Ampofo, G.K.M., Cheng, J., Asante, D.A., Bosah, P., 2020. Total natural resource rents, trade openness and economic growth in the top mineral-rich countries: new evidence from the non-linear and asymmetric analysis. *Resour. Pol.* 68, 101710.
- Asiedu, M., Yeboah, E.N., Boakye, D.O., 2021. Natural resources and the economic growth of West Africa economies. *Appl Econ Finance* 8 (2), 20–32.
- Bansal, M., Kumar, V., 2021. Forcing responsibility? Examining earnings management induced by mandatory corporate social responsibility: evidence from India. *Rev. Account. Finance*.
- Breitung, J., 2001. The local power of some unit root tests for panel data. In: *Nonstationary Panels, Panel Co-integration, and Dynamic Panels*. Emerald Group Publishing Limited.
- Chen, J., 2022. Understanding Dutch Disease, investopedia. <https://www.investopedia.com/terms/d/dutchdisease.asp>. (Accessed 28 September 2022).
- Chen, Z., Umar, M., Su, C.-W., Mirza, N., 2023. Renewable Energy, Credit Portfolios and Intermediation Spread: Evidence from the Banking Sector in BRICS. *Renewable Energy*. <https://doi.org/10.1016/j.renene.2023.03.003>.
- Cheng, C., Ren, X., Wang, Z., 2019. The impact of renewable energy and innovation on carbon emission: an empirical analysis for OECD countries. *Energy Proc.* 158, 3506–3512.
- Deng, W., Akram, R., Mirza, N., 2022. Economic performance and natural resources: evaluating the role of economic risk. *Resour. Pol.* 78, 102840.
- Ekinci, A., 2016. The effect of credit and market risk on bank performance: evidence from Turkey. *Int. J. Econ. Financ. Issues* 6 (2), 427–434.
- Gadea Rivas, M.D., Laeven, L., Pérez-Quiros, G., 2020. Growth-and-Risk Trade-Off.
- Hmaïttane, A., Bouislah, K., M'Zali, B., 2019. Does Corporate Social Responsibility Affect the Cost of Equity in Controversial Industry Sectors? *Review of Accounting and Finance*.
- Huang, Y., Raza, S.M.F., Hanif, I., Alharthi, M., Abbas, Q., Zain-ul-Abidin, S., 2020. The role of forest resources, mineral resources, and oil extraction in the economic progress of developing Asian economies. *Resour. Pol.* 69, 101878.

- Ibrahim, M., Alagidede, P., 2018. Effect of financial development on economic growth in sub-Saharan Africa. *J. Pol. Model.* 40 (6), 1104–1125.
- Ielasi, F., Rossolini, M., Limberti, S., 2018. Sustainability-themed mutual funds: an empirical examination of risk and performance. *J. Risk Finance* 19, 247–261. <https://doi.org/10.1108/JRF-12-2016-0159>.
- Im, K.S., Pasaran, M.H., Shin, Y., 1997. Testing for Unit Roots in Heterogeneous Panels. Faculty of Economics, University of Cambridge.
- Inuwa, N., Adamu, S., Sani, M.B., Modibbo, H.U., 2022. Natural resource and economic growth nexus in Nigeria: a disaggregated approach. *Letters in Spatial and Resource Sciences* 15 (1), 17–37.
- Jarque, C.M., Bera, A.K., 1987. A test for normality of observations and regression residuals. *International Statistical Review/Revue Internationale de Statistique* 163–172.
- Jianu, I., Pircoveanu, L.M., Tudorache, M.D., 2020. The impact of financial risks on economic growth in EU-15. *arXiv preprint arXiv:2008.02166*.
- Jović, S., Maksimović, G., Jovović, D., 2016. Appraisal of natural resources rents and economic development. *Resour. Pol.* 50, 289–291.
- Kao, C., 1999. Spurious regression and residual-based tests for co-integration in panel data. *J. Econom.* 90 (1), 1–44.
- Khan, Z., Hussain, M., Shahbaz, M., Yang, S., Jiao, Z., 2020. Natural resource abundance, technological innovation, and human capital nexus with financial development: a case study of China. *Resour. Pol.* 65, 101585.
- Khan, Z., Ali, S., Dong, K., Li, R.Y.M., 2021. How does fiscal decentralization affect CO₂ emissions? The roles of institutions and human capital. *Energy Econ.* 94, 105060.
- Khan, Z., Hossain, M.R., Badeeb, R.A., Zhang, C., 2023. Aggregate and disaggregate impact of natural resources on economic performance: role of green growth and human capital. *Resour. Pol.* 80, 103103.
- Koenker, R., Bassett Jr., G., 1978. Regression quantiles. *Econometrica. Journal of the Econometric Society*, pp. 33–50.
- Kwakwa, P.A., Adzawla, W., Alhassan, H., Achaamah, A., 2021. Natural resources and economic growth: does political regime matter for Tunisia? *J. Publ. Aff.* e2707.
- Lashitew, A., Werker, E., 2020. Are natural resources a curse, a blessing, or a double-edged sword?, *Brookings*. <http://www.brookings.edu/blog/future-development/2020/07/16/are-natural-resources-a-curse-a-blessing-or-a-double-edged-sword/>. (Accessed 28 September 2022).
- Levin, A., Lin, C.F., Chu, C.S.J., 2002. Unit root tests in panel data: asymptotic and finite-sample properties. *J. Econom.* 108 (1), 1–24.
- Maddala, G.S., Wu, S., 1999. A comparative study of unit root tests with panel data and a new simple test. *Oxf. Bull. Econ. Stat.* 61 (S1), 631–652.
- Mehar, M.R., Hasan, A., Sheikh, M.A., Adeeb, B., 2018. Total natural resources rent relation with economic growth: the case of Pakistan and India. *European Journal of Economic and Business* 3, 03(ISSN-2456-3900).
- Mirza, N., Afzal, A., Umar, M., Skare, M., 2023. The impact of green lending on banking performance: evidence from SME credit portfolios in the BRIC. *Econ. Anal. Pol.* 77, 843–850. <https://doi.org/10.1016/j.eap.2022.12.024>.
- Mohamed, E.S.E., 2020. Resource rents, human development and economic growth in Sudan. *Economies* 8 (4), 99.
- OECD, 2011. The Economic Significance of Natural Resources: Key Points for Reformers in Eastern Europe, Caucasus and Central Asia.
- Ofori, P.E., Grechyna, D., 2021. Remittances, natural resource rent and economic growth in Sub-Saharan Africa. *Cogent Economics & Finance* 9 (1), 1979305.
- Pan, K., Cheng, C., Kirikkaleli, D., Genç, S.Y., 2021. Does financial risk and fiscal decentralization curb the resources curse hypothesis in China? Analyzing the role of globalization. *Resour. Pol.* 72, 102020.
- Phillips, P.C., Perron, P., 1988. Testing for a unit root in time series regression. *Biometrika* 75 (2), 335–346.
- Purewal, K., Haini, H., 2022. Re-examining the effect of financial markets and institutions on economic growth: evidence from the OECD countries. *Econ. Change Restruct.* 55 (1), 311–333.
- Qin, L., Raheem, S., Murshed, M., Miao, X., Khan, Z., Kirikkaleli, D., 2021. Does financial inclusion limit carbon dioxide emissions? Analyzing the role of globalization and renewable electricity output. *Sustain. Dev.* 29 (6), 1138–1154.
- Rahim, S., Murshed, M., Umarbeyli, S., Kirikkaleli, D., Ahmad, M., Tufail, M., Wahab, S., 2021. Do natural resources abundance and human capital development promote economic growth? A study on the resource curse hypothesis in Next Eleven countries. *Resources, Environment and Sustainability* 4, 100018.
- Shabbir, A., Kousar, S., Kousar, F., 2020. The role of natural resources in economic growth: new evidence from Pakistan. *J. Econ. Finance Adm. Sci.* 25 (50), 221–238. <https://doi.org/10.1108/JEFAS-03-2019-0044>.
- Shahbaz, M., Zakaria, M., Shahzad, S.J.H., Mahalik, M.K., 2018. The energy consumption and economic growth nexus in top ten energy-consuming countries: Fresh evidence from using the quantile-on-quantile approach. *Energy Econ.* 71, 282–301.
- Su, C.-W., Rizvi, S.K.A., Naqvi, B., Mirza, N., Umar, M., 2022. COVID19: a blessing in disguise for European stock markets? *Finance Res. Lett.* 49, 103135.
- Umar, M., Ji, X., Mirza, N., Naqvi, B., 2021a. Carbon neutrality, bank lending, and credit risk: evidence from the Eurozone. *J. Environ. Manag.* 296, 113156 <https://doi.org/10.1016/j.jenvman.2021.113156>.
- Umar, M., Ji, X., Mirza, N., Rahat, B., 2021b. The impact of resource curse on banking efficiency: evidence from twelve oil producing countries. *Resour. Pol.* 72, 102080 <https://doi.org/10.1016/j.resourpol.2021.102080>.
- Wen, J., Mughal, N., Kashif, M., Jain, V., Meza, C.S.R., Cong, P.T., 2022. Volatility in natural resources prices and economic performance: evidence from BRICS economies. *Resour. Pol.* 75, 102472.
- Xie, M., Irfan, M., Razaq, A., Dagar, V., 2022. Forest and mineral volatility and economic performance: evidence from frequency domain causality approach for global data. *Resour. Pol.* 76, 102685.
- Yang, J., Rizvi, S.K.A., Tan, Z., Umar, M., Koondhar, M.A., 2021. The competing role of natural gas and oil as fossil fuel and the non-linear dynamics of resource curse in Russia. *Resour. Pol.* 72.
- Yilanci, V., Aslan, M., Ozgur, O., 2021. Disaggregated analysis of the curse of natural resources in most natural resource-abundant countries. *Resour. Pol.* 71, 102017.
- Zhou, H., Li, D., Mustafa, F., Altuntaş, M., 2022. Natural resources volatility and South Asian economies: evaluating the role of COVID-19. *Resour. Pol.* 75, 102524.